BNB CHAIN L2

BEP-20 Native Token: OMC

Omniverse Compute (OMC)

Decentralized GPU Computing Network — Whitepaper Deck v3.0

omc.network · October 2026

Informational only — not an offer to sell or buy any token

EXECUTIVE SUMMARY

Three ideas behind OMC

- Honest scope. Inference, single-node fine-tuning, batch rendering and other embarrassingly parallel jobs — not frontier-scale cluster training.
- Verification you can price. A per-job policy: spot-check re-execution, optional full redundancy, TEE where available, staking and slashing.
- Token demand tied to usage. Fees paid per job, part of every fee buys back and burns OMC, rewards are paid only against verified work.

Layer	Role
L1 — BNB Chain	OMC token, staking, DAO governance, final arbitration
L2 — OP Stack (opBNB)	Heartbeats, job records, state-channel settlement, micropayments
Off-chain grid	Sandboxed containers with CUDA / ROCm acceleration

PROBLEM

Compute is scarce, expensive and centralized

Demand

AI developers face high hourly prices, regional quota limits and vendor lock-in; studios face sharp render peaks.

Supply

Consumer GPUs, small data-center idle capacity and post-PoW hardware sit underused.

The gap

No widely trusted way for strangers to rent GPU time with guarantees about correctness, privacy and payment.

The answer

OMC: a DePIN market with verifiable jobs, USD-denominated staking and per-job settlement on BNB Chain.

TARGET USERS

Who runs what on OMC

AI startups & developers

Inference, LoRA/QLoRA fine-tuning, batch embeddings — lower cost, no quota waits.

Web3 & DeAI projects

Agent inference and batch workloads with crypto-native payment.

Rendering & 3D studios

Blender and UE offline rendering with per-job pricing and burst capacity.

GPU owners & small DCs

Monetize idle hardware with predictable earnings and low operating friction.

ARCHITECTURE

Scheduling in two stages

- Stage 1 — hard filters: VRAM, driver and CUDA/ROCm version, region, confidential-computing requirement, minimum stake and reputation.
- Stage 2 — scoring: each metric becomes a percentile rank among eligible nodes, then a weighted matching problem maximizes total score; the final pick is drawn at random from the top-k candidates.

Job class	TFLOPS	VRAM	Bandwidth	Latency	Reputation
Batch and rendering (proposed)	0.30	0.20	0.15	0.10	0.25
Inference (proposed)	0.20	0.15	0.10	0.30	0.25

VERIFICATION AND TRUST

No single technique verifies every workload

Mechanism	How it works	Cost to requester
Spot-check re-execution	A random share of jobs is silently re-run on an independent node	Base × (1 + a); proposed a = 5–10%
Full redundancy	The job runs on 2–3 independent nodes; outputs are compared	2× to 3×
Confidential computing (TEE)	Hardware-protected execution with remote attestation	Premium for TEE nodes
Zero-knowledge proofs	Research track — not offered at launch	—

Non-determinism is handled with task-specific rules: exact hashes, numerical tolerance, or fixed seeds.

SECURITY AND PRIVACY

Honest about what cannot be guaranteed

- Isolation: hardened sandboxes (gVisor by default, Kata / microVM tiers for sensitive workloads), no host access, no outbound network by default.
- Encryption: inputs, outputs and images encrypted in transit and at rest; keys released only after assignment.
- Three data-protection levels: Standard, Protected (ephemeral, wiped), Confidential (TEE-only).
- Honest limitation: on non-TEE nodes a malicious provider could observe data in GPU memory — sensitive workloads should choose the Confidential level.
- Assurance: audits before mainnet with public reports; bug bounty from mainnet (critical findings from $50,000, proposed).

ECONOMICS

Fee flows tied to real usage

Flow	Split
Protocol fee at settlement	3% of every job
Buy back and burn	30% of the fee = 0.9% of settled volume
Reputation Reward Pool	70% of the fee, paid for verified work
Payments	OMC (10% fee discount, proposed) or USDT / USDC on BNB Chain
Emission scaling (proposed)	daily emission = ceiling × min(1, U / Utarget), Utarget = 60%

0.9%

of settled volume is burned

3% fee × 30% burn share

$10M

monthly volume → ~$90,000 burn

illustration only, not a forecast

TOKEN DISTRIBUTION

1,000,000,000 OMC fixed supply (BEP-20)

Category	Share	Amount	Vesting
Compute mining and node rewards	50%	500,000,000	0% at TGE; 10-year emission, halving every 2 years
Institutional investors	10%	100,000,000	5% at TGE, 6m cliff, 24m monthly
Private investors	8%	80,000,000	10% at TGE, 3m cliff, 18m monthly
Community airdrop	2%	20,000,000	100% at TGE
Ecosystem and grants	15%	150,000,000	5% at TGE, 6m lock, 36m monthly
Team and core contributors	10%	100,000,000	0% at TGE, 12m cliff, 24m monthly
Treasury and liquidity	5%	50,000,000	50% at TGE for liquidity; rest by DAO Treasury

Circulating supply at TGE ≈ 65.5M OMC (6.55%). A public vesting dashboard shows every category.

Airdrop is a fixed 1,000,000-entry pool (20 OMC each): up to 5 campaign entries + up to 5 referral entries per wallet.

EMISSION

A ceiling, not an entitlement

Schedule	Annual ceiling	Share of pool
Years 1–10, flat — no halving, no epochs	50,000,000 OMC	10.0% every year

- 500,000,000 OMC released in 10 equal annual instalments, linear by day.
- Rewards are paid only against verified work; daily emission scales with utilisation.
- Unused emission goes to an Emission Reserve — burned or re-released only by DAO vote.

STAKING AND SLASHING

Five hardware tiers, penalties paid to the wronged

Tier	Minimum stake (OMC, proposed)	Offense	Penalty
1 · consumer GPU	20	Job timeout	2%
2 · workstation GPU	100	Offline during a job	5%
3 · 80 GB data-center GPU	500	Incorrect output	30%
4 · multi-GPU node	1,000	Proven fraud	100%
5 · cluster-scale	5,000	—	—

- Stake is a published OMC ladder; a wallet that completes the airdrop reaches tier 2 without buying tokens.
- Amounts are proposed and adjustable by governance before mainnet.

Slashed tokens: 50% compensate requesters · 30% DAO Treasury · 20% burned.

GOVERNANCE AND ROADMAP

From team-led to community-led

- Launch: 4-of-7 multisig of core contributors with a 3-day timelock (proposed); signers not disclosed at this stage.
- Transition: OMC holders vote on parameters and Treasury spending — proposed: 100,000 OMC threshold, 10% quorum, 7-day voting, 2-day delay.
- Full DAO (2027 or later): multisig removed or limited to emergency actions.
- Phase 1 (2026 Q4, in progress): testnet on BNB Chain (Chapel), token contract, open-source node client; the 2% community airdrop opens January 1, 2027.
- Phase 2 (2027 Q1, planned): mainnet, emission start, airdrop claim window and staking all open on January 1, 2027; Hugging Face and Blender integrations, completed audits.
- Phase 3 (2027 Q2 and later): TEE tier, ZK research results, DAO transition, cross-chain settlement.

LEARN MORE

Where to verify everything

Website

https://omc.network — news, whitepaper v3.0, testnet, airdrop

DCF forum

https://forum.omc.network — GPU pricing research and deep dives

Repository

https://github.com/CBTdao/omc-network — open-source node client

Contact

support@omc.network · security@omc.network

Third-party market figures are sourced and dated in Whitepaper v3.0, Section 13.